# Dr. Romain Beucher


## Geoscience Modelling & Software Engineering Expert

Thermo-Mechanical Modelling | Surface Processes | Research Software Development

Canberra, Australian Capital Territory, Australia



**Professional Summary:** Dr. Romain Beucher is a geoscientist and research software engineer with extensive expertise in lithospheric-scale thermo-mechanical modelling, surface processes, and low-temperature thermochronology. With over 15 years of research experience across leading institutions in Australia, Norway, and the UK, he specializes in coupling large-scale basin models with surface processes to study interactions between erosion and tectonics. Currently leading the Model Evaluation and Diagnostics team for the Australian Earth-System Simulator National Research Infrastructure (ACCESS-NRI), Dr. Beucher combines deep geological knowledge with advanced software engineering capabilities.


## Professional Experience

### Team Leader (Senior Manager) – Model Evaluation and Diagnostics

**ACCESS-NRI (Australian Earth-System Simulator National Research Infrastructure)**

*August 2022 – Present (3+ years) | Canberra, ACT, Australia*

Leading the development and evaluation of model diagnostics for Australia's national Earth-system modelling infrastructure.

**Key Skills:** Negotiation, Training, Team Management, Release Management, Robust Design, Critical Thinking, Problem Solving, Teamwork

### Researcher

**The Australian National University (ANU)**

*February 2020 – August 2022 (2 years 7 months) | Canberra, ACT, Australia*

Conducted advanced research in lithospheric-scale thermo-mechanical modeling of rifts and passive margins. Specialized in surface processes modeling with focus on quantifying rock exhumation and relief evolution using low-temperature thermochronology (apatite Fission Track, U-Th/He). Developed coupled large-scale basin models integrating surface processes to study erosion-tectonic interactions and feedbacks.

**Key Skills:** Training, Release Management, Robust Design, Critical Thinking, Problem Solving, Teamwork

### Research Software Engineer

**AuScope**

*October 2019 – August 2022 (2 years 11 months)*

Developed research software infrastructure for geoscience applications.

**Key Skills:** Training, Release Management, Robust Design, Critical Thinking, Problem Solving, Teamwork

### Visiting Research Fellow

**The Australian National University (ANU)**

*October 2019 – February 2020 (5 months) | Canberra, Australia*

**Key Skills:** Critical Thinking, Problem Solving

### Postdoctoral Researcher

**University of Melbourne**

*October 2016 – February 2020 (3 years 5 months) | Melbourne, Australia*

**Key Skills:** Critical Thinking, Problem Solving

### Postdoctoral Researcher

**University of Bergen (UiB)**

*June 2013 – June 2016 (3 years 1 month) | Bergen Area, Norway*

**Key Skills:** Critical Thinking, Problem Solving

### Research Associate

**University of Glasgow**

*June 2010 – June 2013 (3 years 1 month) | Glasgow, United Kingdom*

**Key Skills:** Critical Thinking, Problem Solving

## Education

### Doctor of Philosophy (PhD) – Geology/Earth Science

**Université Grenoble Alpes**

*2006 – 2009*

### Master of Science (MS) – Geology/Earth Science

**Université Grenoble Alpes**

*2004 – 2006*

### Bachelor of Science (BS) – Geology/Earth Science

**Le Mans Université**

*2001 – 2004*

## Core Competencies

- Lithospheric-scale thermo-mechanical modelling of rifts and passive margins
- Surface processes modeling and geomorphological evolution
- Low-temperature thermochronology (apatite Fission Track, U-Th/He dating)
- Quantitative analysis of rock exhumation and relief evolution
- Coupled basin-scale models integrating tectonics and erosion processes
- Research software engineering and model development
- Earth-system model evaluation and diagnostics
- Team leadership and research project management